

Opinion Network Formation: How This Class’s Views on Technology, Education, Society, and the Environment Fit Together

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Repository: https://github.com/shrey715/dpcn-opinion_network-asstra

1 Dataset Documentation

The survey collected 96 responses across 60 Likert-scale statements, split into four blocks: Technology (T01–T15), Education (E01–E15), Society and Ethics (S01–S15), and Environment and Values (V01–V15). We mapped Strongly Disagree through Strongly Agree to the integers 1 through 5. A few respondents answered “No Comments” on individual items instead of picking a point on the scale; we coded those as missing rather than forcing them to the midpoint, since a shrug isn’t the same as “Neutral.”

Respondent filtering. Five people left the whole survey blank and got dropped outright. Six more stopped partway through, and not randomly: each answered items $1, \dots, N$ in order and then left everything after that blank. That looks like fatigue, not scattered skipping. A further 17 people finished the survey but marked “No Comments” on one to four items scattered through it, so those got coded as missing too. That’s the full breakdown of all 96: 5 blank, 6 partial with a clean cutoff, 17 partial with scattered gaps, and 68 who answered everything. Every analysis that needs a complete numeric matrix uses only those 68, so the partial responses can’t skew a correlation.

Descriptive summary. Before building any network, it’s worth checking what the class agreed on and what it fought about, since that shapes what a correlation network can even find. The numbers below come from the 91 non-blank respondents (all 96 minus the 5 blank surveys), not the 68 used everywhere from Section 2 on; the two samples give slightly different numbers, and these are the 91-respondent ones. Environment has the highest mean (4.39 out of 5) and the tightest spread of any block (mean per-item standard deviation 0.72): pretty much nobody disagrees that companies should answer for their environmental impact. Education has the lowest mean (3.91), and its item E03 (“class attendance should be compulsory”) is the single most contested statement in the whole survey, mean 2.06, standard deviation 1.12. By spread rather than mean, though, Technology is actually the most divided block (0.92), just ahead of Education (0.89), so Education has the lowest mean but not the widest spread. This consensus-versus-conflict split shows up again, more sharply, once we look at the data as a network instead of a table of averages (Section 4), though Section 4 also shows part of that network-level split comes from a general agreement factor rather than topic content alone.

2 Pipeline Followed

We built four networks from the same cleaned data, each aimed at a different question about how the class’s opinions fit together. In every case an edge, whether between two statements or two respondents, means their answers move together across the sample: a Pearson correlation, weighted by its strength. Table 1 summarizes all four before we walk through each one.

Table 1: The four network constructions at a glance.

Network	Nodes	Edge rule	Validation
Idea	60 statements	$ r \geq 0.15$, weighted	permutation null ($B=1000$)
Extension 1	60 statements	unsigned, $\tau \in [0.10, 0.45]$	Erdos-Renyi null ($B=200$)
Extension 2	68 respondents	5-NN, profile correlation	leave-one-item-out + bootstrap
Extension 3	60 statements (4 blocks)	within-block $ r $, unthresholded	permutation null ($B=200$, per block)

2.1 Idea: the statement co-endorsement network

The 60 statements are the nodes. Two statements get an edge if the absolute Pearson correlation between their answers, across the 68 complete respondents, is at least 0.15, weighted by that correlation. That gives 907 edges on 60 nodes, a deliberately dense graph. Cutting down to a sparse top- $N\%$ graph would just bake in an arbitrary cutoff; keeping the full graph means centrality reflects the whole correlation structure instead of whatever a cutoff happened to keep. On this graph we ran eigenvector centrality and PageRank, which both find statements near the core of the belief system, plus betweenness centrality for statements that bridge otherwise separate clusters. We also ran k -core decomposition on a stricter $|r| \geq 0.30$ version to find a backbone: the subgraph left after repeatedly stripping out any node connected to fewer than k others in it.

We checked significance with a permutation test: shuffle each of the 60 item columns independently across respondents, 1000 times, which wipes out any real correlation but keeps each item’s own response distribution intact. Then we compare the real statistic to that null distribution. Raw Pearson correlation is also sensitive to how agreeable a respondent is overall, so we separately reran the whole thing on a version of the data where each respondent’s own mean response is subtracted out first. Both readings are discussed together in Section 4.1.

2.2 Extension 1: structure and sensitivity of the statement network

This part asks four questions about the statement network’s structure, all on the same 60 nodes. **1a** checks whether the graph at $|r| \geq 0.20$ (647 edges: 620 positive, 27 negative) is more triangle-dense than a random graph of the same size and density would be. We compared the observed triangle count to an Erdos-Renyi graph $G(60, \hat{p})$, both through the closed form $\binom{60}{3} \hat{p}^3$ and through 200 simulated draws. **1b** pulls the 27 negative edges out and looks at them as their own subgraph: degree, connected components, and whether any triangles form among purely negative relationships. **1c** sweeps the edge threshold τ from 0.10 to 0.45 and tracks edge count, giant-component size, and isolated-node count against their own nulls (a permutation null for the first two, the Erdos-Renyi formula $n(1 - \hat{p})^{n-1}$ for isolated nodes), plus the Technology-versus-rest degree gap at every threshold. **1d** builds a structural table, density, mean degree, degree variance, components, diameter, average path length, clustering, and degree-weighted average-neighbor-degree at $|r| \geq 0.15, 0.20, 0.30$, against the matching Erdos-Renyi predictions, plus the degree distribution at $|r| \geq 0.30$ against a Binomial($n-1, p$) null. 1d also answers a question the Idea section leaves open: why 0.15? A permutation null tells us what fraction of edges at each threshold is there by chance alone.

2.3 Extension 2: the respondent similarity network

Here the 68 respondents are the nodes, each linked to its five nearest neighbors by profile correlation. We used this network to predict every one of the $68 \times 60 = 4080$ (respondent, item) answers from a respondent’s neighbors, and compared the error to a global item-mean baseline. Similarity for predicting a given item is recomputed with that item’s own column dropped first, so a respondent’s real answer never sneaks into the neighbor selection used to predict it. Since the 4080 cells come from only 68 people, we used a respondent-level bootstrap instead of a cell-level test that would wrongly treat correlated cells as independent. We also draw the network itself as a directed graph (Figure 7, since each respondent points to five others but isn’t necessarily pointed back at by the same five) along with its in-degree distribution.

2.4 Extension 3: per-block network density

The 60 statements split into four blocks, T, E, S, V, each treated as its own induced subgraph of the full statement-correlation graph, with no edges across blocks. Each block’s weighted link density is the mean of $|r|$ over every pair inside it, straight off the correlation matrix. We test that density against a permutation null built the same way as Section 2.1’s, run separately per block, and check it against the closed-form expected value of $|r|$ for two independent 68-observation vectors. Like the Idea network, we reran this on a row-centered correlation matrix as a robustness check (Section 4.4).

3 Analysis and Visualizations

The figures and tables below cover all four networks and their significance tests, in the same order as Section 2. Section 4 walks through what each one means.

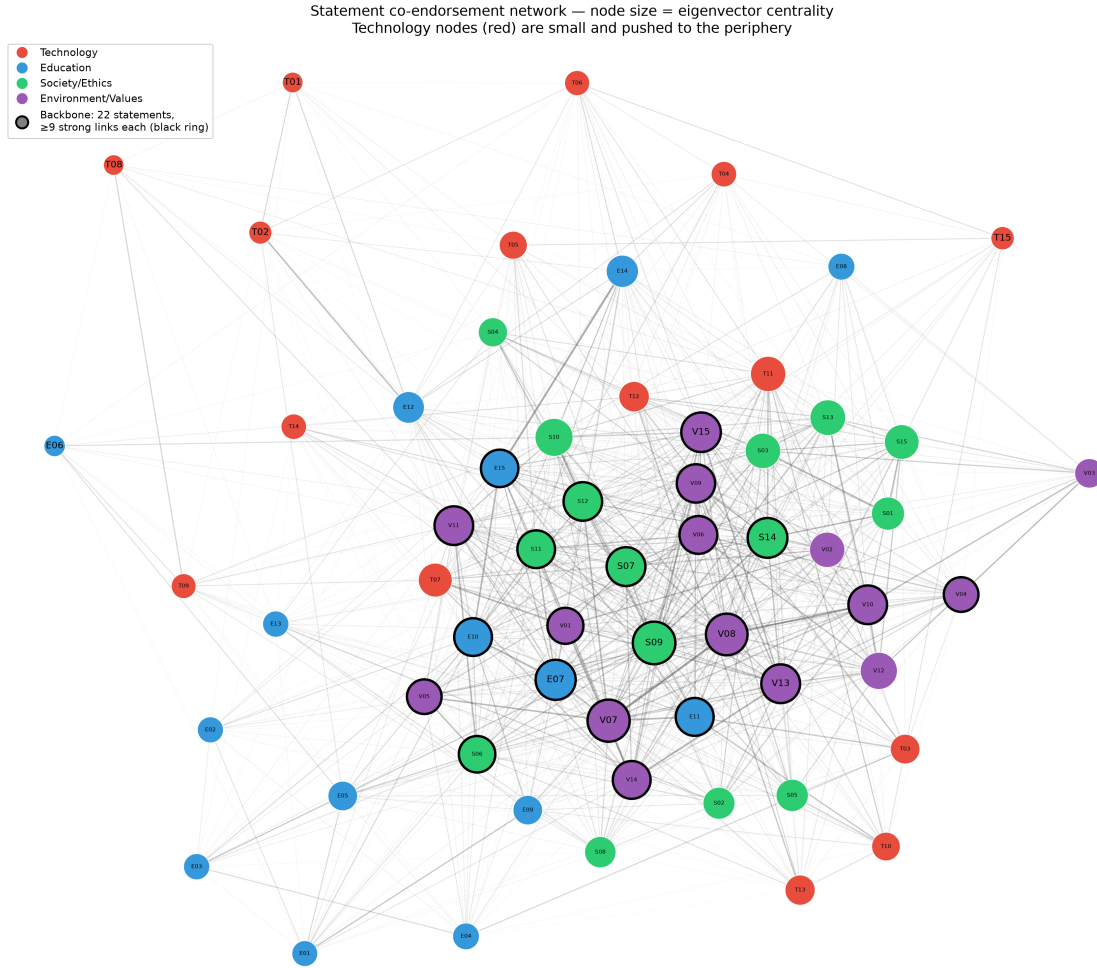


Figure 1: The statement co-endorsement network. Node size is eigenvector centrality, color is the T/E/S/V block, a black ring marks the 22-statement backbone. Technology nodes are small and sit toward the edge.

4 Results and Discussion

4.1 Idea: a decoupled Technology cluster, with a caveat

In the raw data, the centrality ranking puts statements about inclusivity, environmental accountability, and educational reform at the top, and Technology statements at the bottom almost across the board (Table 5). The 22-statement backbone, the core where every member connects strongly to at least nine others, doesn’t

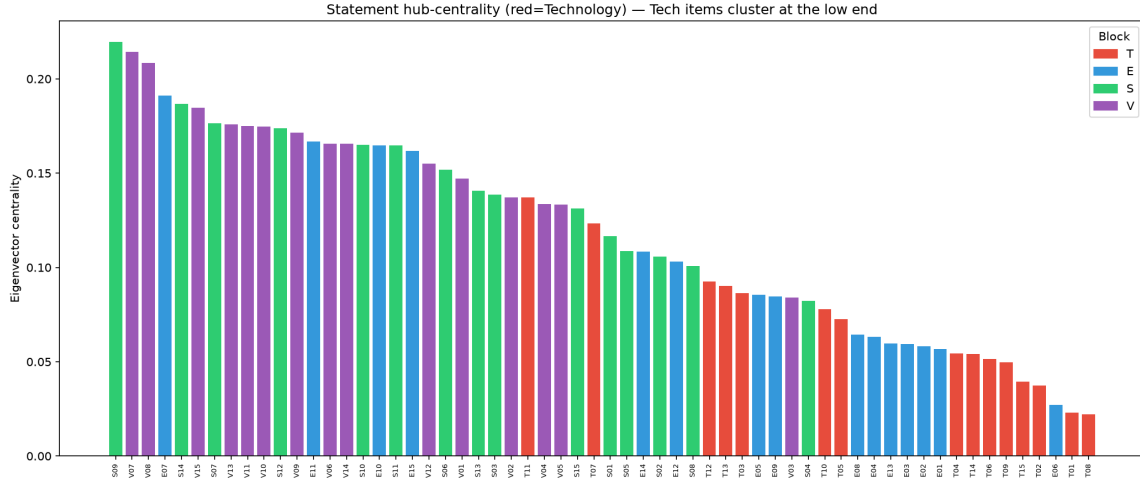


Figure 2: Eigenvector centrality for all 60 statements, sorted and colored by block. Technology clusters at the low end.

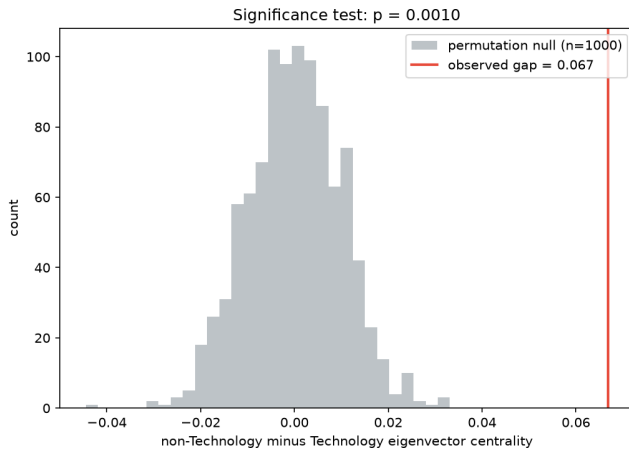


Figure 3: Permutation test for the centrality gap: the real value (red line) against 1000 reshuffled versions of the data (gray bars).

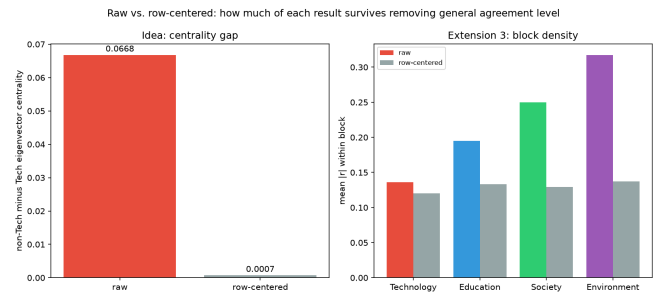


Figure 4: Raw versus row-centered: the centrality gap and the four block densities, before and after subtracting each respondent's general agreement level. Both shrink a lot.

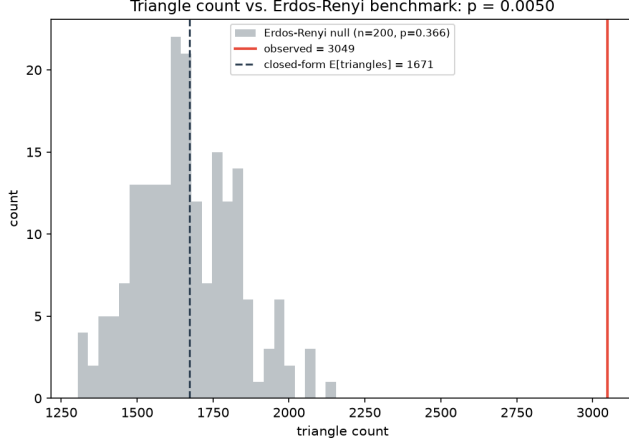


Figure 5: 1a: observed triangle count in the 647-edge graph (red line) against 200 simulated Erdos-Renyi graphs of the same size and density (gray bars).

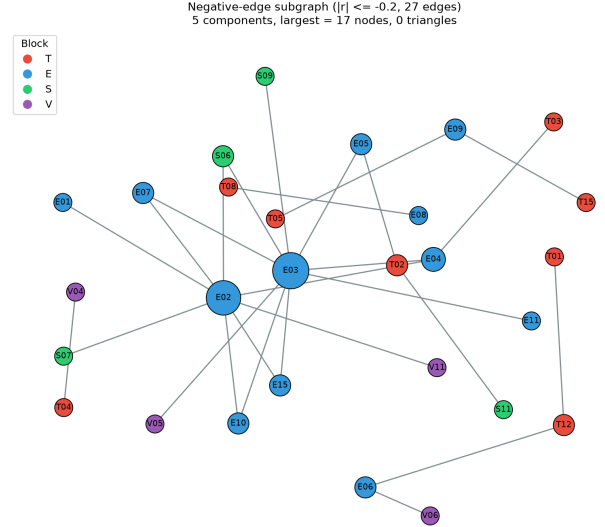


Figure 6: 1b: the 27 negative edges as their own subgraph. Node size is degree within this subgraph; E03 and E02 (both Education) anchor the biggest component.

Table 2: 1c: threshold sweep over $\tau \in [0.10, 0.45]$. Edges and giant-component size are shown observed vs. permutation null; isolated nodes are observed vs. the Erdos-Renyi prediction $n(1 - \hat{p})^{n-1}$; the last column is the Technology-versus-rest mean-degree gap.

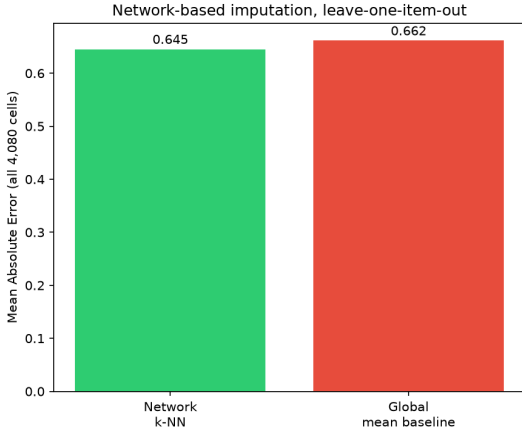
τ	edges (obs)	edges (null)	giant (obs)	giant (null)	isolated (obs)	isolated (ER)	Tech gap
0.100	1172	734.1	60	60.0	0	0.00	9.956
0.125	1029	554.9	60	60.0	0	0.00	11.600
0.150	907	399.1	60	60.0	0	0.00	11.689
0.175	783	272.6	60	60.0	0	0.00	11.511
0.200	647	177.5	60	59.9	0	0.00	11.956
0.225	565	115.2	60	58.8	0	0.00	11.244
0.250	457	68.5	60	51.0	0	0.00	9.822
0.275	368	37.5	60	21.6	0	0.00	8.089
0.300	304	23.6	60	9.6	0	0.00	6.844
0.325	233	11.8	56	3.8	2	0.01	6.178
0.350	171	6.4	47	3.1	8	0.15	5.289
0.375	125	3.7	44	2.5	12	0.80	3.956
0.400	95	1.6	42	2.0	16	2.32	3.067
0.425	59	0.6	35	1.4	19	8.12	2.000
0.450	41	0.2	33	1.2	21	15.05	1.200

Table 3: 1d: structural stats at three thresholds, observed value with the Erdos-Renyi prediction in parentheses.

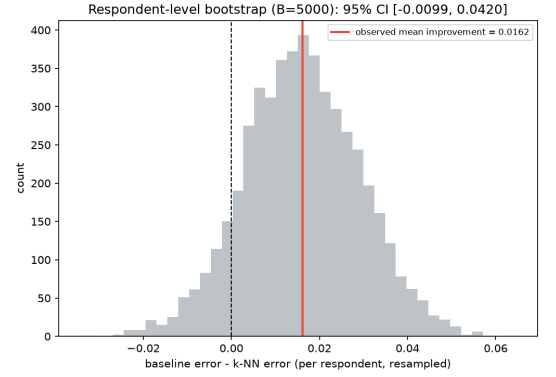
τ	edges	density	$\langle k \rangle$	diameter	Var(k)	$\langle L \rangle$	$C / \langle k_{nn} \rangle$
0.15	907	0.512	30.2	2	94.3 (15.0)	1.49 (1.20)	0.64 (0.51) / 33.4 (30.7)
0.20	647	0.366	21.6	3	95.0 (13.9)	1.67 (1.33)	0.57 (0.37) / 26.0 (22.2)
0.30	304	0.172	10.1	5	45.2 (8.5)	2.21 (1.77)	0.45 (0.17) / 14.6 (11.0)

Table 4: One headline statistic per analysis, tested against its own null.

Analysis	Statistic	Observed	Test
Idea (raw)	Eigenvector centrality gap, non-Tech – Tech	0.0668	$p = 0.0010$ ($B=1000$)
Idea (row-centered)	Same statistic, agreement level removed	0.0007	near null, see §4.1
Ext. 1a	Triangle count, 647-edge graph	3049	$p = 0.0050$ vs. ER null
Ext. 1b	Negative-edge subgraph	hub-and-spoke on E03/E02, 0 triangles	descriptive
Ext. 1d	Clustering at $\tau = 0.30$, obs vs. ER	0.45 vs. 0.17	Table 3
Ext. 2	Network k -NN MAE / baseline (4080 cells)	0.645 / 0.662	95% CI $[-0.010, 0.042]$
Ext. 3	Within-block mean $ r $, T/E/S/V	0.136/0.195/0.250/0.318	$p = 0.0050$ (all four)



MAE over all 4080 cells.



Respondent-level bootstrap; the 95% CI includes zero.

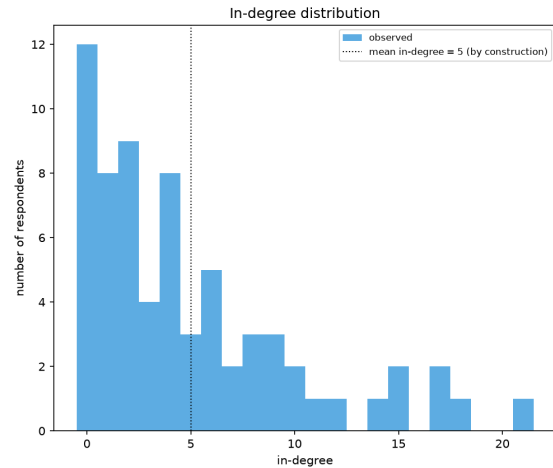
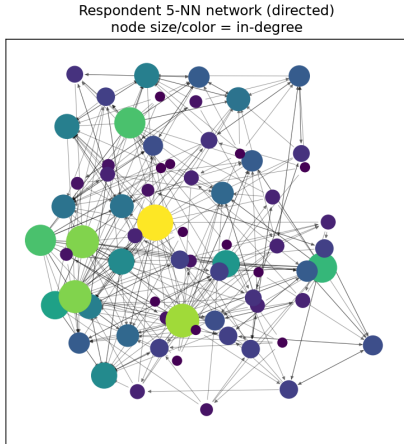


Figure 7: Top: network k -NN against the global-mean baseline, and the bootstrap behind the confidence interval in Table 4. Bottom: the directed 5-NN respondent network and its in-degree distribution, a handful of respondents get picked as a neighbor far more often than random 5-NN assignment would predict.

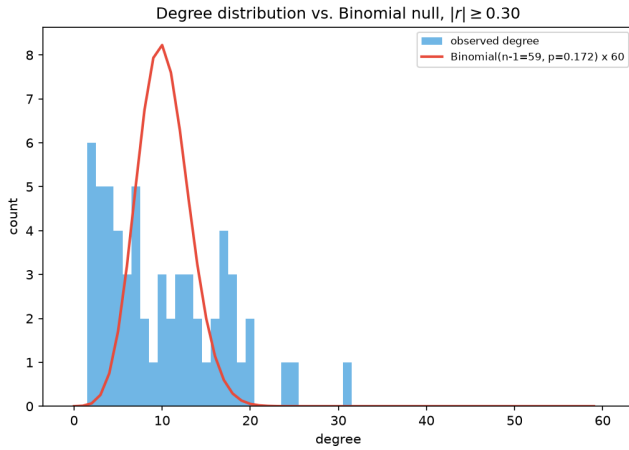


Figure 8: 1d: degree distribution at $|r| \geq 0.30$ (blue bars) against a $\text{Binomial}(59, \hat{p})$ null of the same mean (red line). The real distribution is wider.

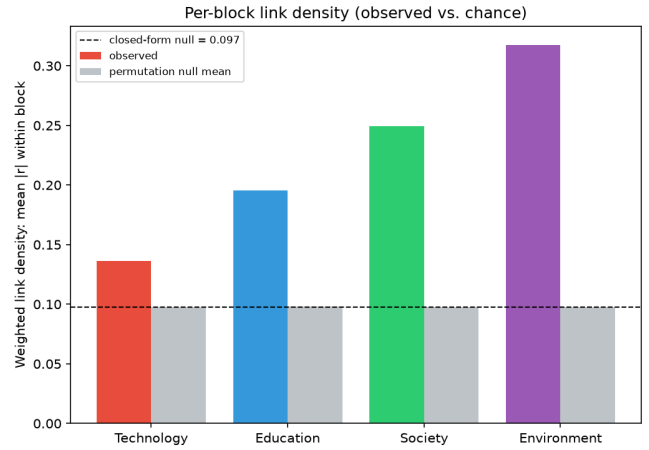


Figure 9: Within-block link density by topic, against each block’s own permutation null and the closed-form null $\sqrt{2/(\pi(n-1))} \approx 0.098$ (dashed line).

contain a single Technology statement. We tested the Technology-versus-rest centrality gap against 1000 independent reshuffles and never once saw it matched by chance ($p = 0.0010$), putting the real value about seven standard deviations above the null mean. PageRank tells the same story as eigenvector centrality (the two rankings correlate at 0.985, so PageRank isn’t adding new information here), and betweenness centrality shows the same pattern: Technology statements bridge the network less than everything else does.

That gap isn’t only about Technology as a topic, though. Pearson correlation is sensitive to how agreeable a respondent is in general, and this class’s average per-item agreement ranges from 3.27 to 4.77 out of 5 depending on who you ask. When we rebuild the same graph after subtracting each respondent’s own mean response first, which removes that general agreement level before correlating items, the centrality gap drops from 0.0668 to 0.0007, basically zero (Figure 4). That doesn’t prove the raw finding is fake. A shared leaning toward pro-social, pro-environment statements could be a real attitude and not just survey noise, and stripping it out would erase real signal if that’s the case. But we can’t tell those two stories apart with this data, so both need reporting: Technology attitudes are less central in the raw numbers, and most of that gap tracks general agreement rather than Technology specifically.

4.2 Extension 1: structure, disagreement, and threshold sensitivity

1a, triangle density. The 647-edge graph has 3049 triangles and a transitivity (clustering coefficient) of 0.566, against a mean of 1663 triangles and 0.364 transitivity across 200 simulated Erdos-Renyi graphs of the same size and density ($p = 0.0050$; Figure 5). So the graph is more triangle-dense than random chance would give it. That’s expected for any correlation graph: if statement A and B both correlate with C, A and B tend to correlate with each other too, through that shared variance. It’s a sanity check on the graph’s structure, not proof of some deeper psychological consistency mechanism.

1b, the negative-edge subgraph. Pulling out just the 27 negative edges gives five small components (sizes 17, 4, 3, 2, 2) with zero triangles (Figure 6). The biggest one is a hub-and-spoke shape built around two Education items: E03 (“attendance should be compulsory,” degree 9) and E02 (“traditional exams accurately measure knowledge,” degree 8). They aren’t directly connected to each other, but they each disagree with a lot of the same surrounding statements. So the disagreement here is centered on two contested Education items, not on Technology.

1c, threshold sweep. Sweeping τ from 0.10 to 0.45 (Table 2) shows the null graph’s giant component falling apart well before the real one’s does: the permutation null starts shrinking hard around $\tau \approx 0.275$ –0.30, while the real giant component holds all 60 nodes through $\tau = 0.30$ and only starts losing them after

Table 5: The five most and least central statements by eigenvector centrality. Four of the five least central are Technology items.

Code	Statement	Centr.
<i>Most central (top 5 of 60)</i>		
S09	Universities should promote an inclusive environment where different viewpoints can be discussed respectfully.	0.220
V07	Protecting biodiversity is essential for long-term human well-being.	0.214
V08	Universities should adopt environmentally sustainable campus practices even if implementation costs increase.	0.208
E07	Publishing research before graduation should be encouraged, but not mandatory.	0.191
S14	Leaders should prioritize ethical decision-making even when it reduces short-term gains.	0.187
<i>Least central (bottom 5 of 60)</i>		
T08	AI-assisted diagnosis should become routine in healthcare.	0.022
T01	Artificial Intelligence will improve society more than it will create problems.	0.023
E06	Every undergraduate student should participate in at least one research project.	0.027
T02	Generative AI tools should be allowed as learning aids in higher education.	0.037
T15	Society will increasingly depend on AI-assisted decision making.	0.039

that. Going from $\tau = 0.30$ to $\tau = 0.35$, 13 statements fall out of the giant component, and 9 of them are Technology. Isolated nodes show up in much bigger numbers than Erdos-Renyi predicts once the threshold gets strict enough to break the graph apart (16 observed against 2.3 expected at $\tau = 0.40$). That fits a graph built from real, uneven correlations rather than uniformly random ones: a tight, well-connected core hangs together while the weaker, peripheral statements peel off first. The Technology-versus-rest degree gap stays positive at every single threshold from 0.10 to 0.45, which is why using $\tau = 0.15$ for the Idea network is a reasonable choice: the Technology finding doesn’t depend on this specific number.

1d, structural comparison and degree distribution. Table 3 lines the graph up against Erdos-Renyi at $\tau = 0.15, 0.20, 0.30$. At every threshold, the observed clustering, degree variance, and average-neighbor-degree all beat the Erdos-Renyi prediction for a random graph of the same size and density, most obviously at $\tau = 0.30$ (clustering 0.45 versus a predicted 0.17; degree variance 45 versus a predicted 8.5). The degree distribution at $\tau = 0.30$ is noticeably wider than a Binomial($59, \hat{p}$) null with the same mean (Figure 8). This same permutation approach also answers why the Idea network uses $|r| \geq 0.15$: roughly 43% of its 907 edges (about 393) would show up by chance alone, falling to about 28% at $\tau = 0.20$ (179 of 647) and about 7% at $\tau = 0.30$ (23 of 304). We chose 0.15 on purpose, accepting more chance-level edges in exchange for a graph whose centrality reflects the full correlation structure instead of an arbitrarily thin slice of it. Part 1c shows this choice doesn’t drive the Technology result, it just controls how much noise rides along with the real signal.

4.3 Extension 2: a small, borderline predictive effect

Across all 4080 (respondent, item) cells, with each item’s own column dropped from the similarity calculation before predicting it, the network’s k -NN prediction gets a mean absolute error of 0.645 against a global-mean baseline of 0.662. That’s a small improvement. Because those 4080 cells come from only 68 people and aren’t independent of each other, we used a respondent-level bootstrap instead of a cell-level test: the 95% confidence interval for the improvement is $[-0.010, 0.042]$ (Figure 7), which includes zero. The honest read is a small, borderline effect, not a strong one.

Looking at the 5-nearest-neighbor structure as a directed graph (Figure 7, since each respondent points to five others but isn’t necessarily pointed back at by the same five) shows the in-degree distribution is way more uneven than random assignment would give: standard deviation 5.07 against 2.14 under random

assignment, a max in-degree of 21, and 12 respondents nobody picked at all. In-degree correlates with a respondent’s own mean agreement level ($r = 0.43$), people who agree with more statements on average get picked as a neighbor more often. We don’t know why that correlation holds; it’s reported here as a descriptive fact, not an explained one. The six respondents who abandoned the survey partway through are excluded before this network gets built, so this analysis says nothing about them.

4.4 Extension 3: topic-level density, with the same caveat as the Idea

Every block clears its own permutation null (Table 4), and the empirical null mean of about 0.098 lines up with the closed-form expectation for two independent 68-observation vectors, $\sqrt{2/(\pi \cdot 67)} \approx 0.0975$. But the effect sizes are pretty different: Environment’s density above its noise floor (mean $|r| = 0.318$) is roughly 5.7 times Technology’s (mean $|r| = 0.136$). Just like the Idea network, this gap shrinks a lot once each respondent’s general agreement level gets removed first: the row-centered densities converge to roughly 0.12–0.14 across all four blocks instead of spreading from 0.136 to 0.318 (Figure 4). Technology’s within-block correlations are weakly related to each other compared to Environment’s; nothing here tests for, or backs up, separate sub-views inside any one block.

4.5 Synthesis

These four analyses are several views of one correlation structure, not four fully independent confirmations of each other. The Idea, Extension 1, and Extension 3 all come from the same 60-by-60 item correlation matrix; only Extension 2 uses a genuinely different one, built respondent by respondent. Even so, read together they point the same way. In the raw data, Technology and AI statements are less central, they’re behind the graph’s only real disagreement structure once Education’s hub items are set aside, and they form the least internally correlated topic block. But the row-centered checks in Sections 4.1 and 4.4 show most of that pattern tracks a general agreement factor shared across respondents, not something specific to Technology, and Extension 2’s corrected result is small and statistically borderline rather than strong. We didn’t test, and this report doesn’t claim, that someone’s environmental or ethical positions predict their stance on AI; that would need its own targeted analysis that this project doesn’t include.

5 Individual Contribution

- Abhishek** Built the shared data-loading and permutation-null utility (`data_utils.py`) that the rest of the project runs on. Built the statement co-endorsement network and computed its eigenvector centrality, PageRank, betweenness centrality, and k -core backbone. Ran the centrality-gap significance test at $B = 1000$, the PageRank-versus-eigenvector check, the per-block betweenness stats, and the row-centered robustness check in Section 4.1. Produced the network visualization in Figure 1 along with its centrality and significance plots.
- Shreyas** Built Extension 1: the Erdos-Renyi triangle-density benchmark, the negative-edge subgraph analysis, the threshold sweep across edge count, giant-component size, isolated nodes, and the Technology degree gap, and the structural comparison table with the degree-distribution check against a Binomial null, including the chance-edge-fraction argument for why the Idea network uses $|r| \geq 0.15$. Produced Figures 5, 6, and 8, and Tables 2 and 3.
- Sampath** Built the leave-one-item-out evaluation of the respondent similarity network across all 4080 cells, the respondent-level bootstrap behind its significance test, and the directed-graph in-degree analysis and visualization of the 5-NN respondent network. Built Extension 3’s closed-form null derivation, its weighted-link-density framing, and its row-centered robustness check, and put together the combined raw-versus-row-centered comparison figure covering both the Idea and Extension 3 results. Produced Figures 4, 7, and 9.

All three of us checked each other’s numbers against the shared repository before putting this report together, and wrote the synthesis in Section 4.5 together.